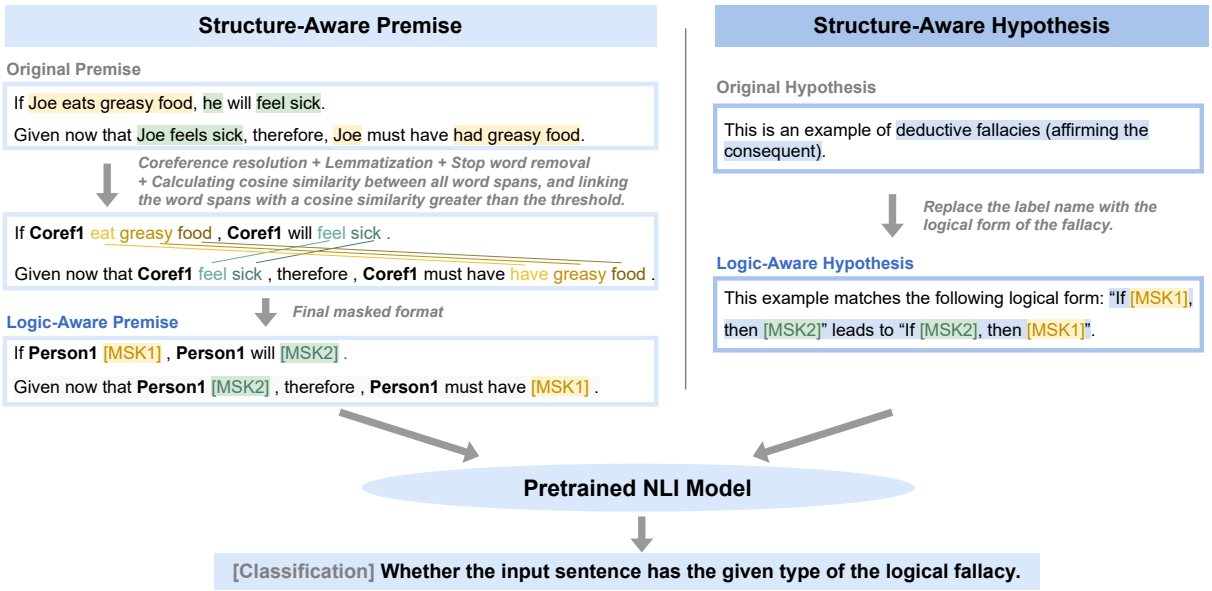

Figure 2: Our baseline model is a structure-aware classifier based on pretrained NLI model, with a structure-aware premise and structure-aware hypothesis. The structure-aware premise masks the content words to distill the argument structure. Specifically, we first resolve the coreferences, and then apply Sentence-BERT to match the lemmatized word spans (excluding the stopwords) whose contextualized embeddings have a cosine similarity larger than a certain threshold. And the structure-aware hypothesis uses the standard logical form of the given fallacy type.

Logical Fallacy Type	Frequency in Data
Intentional Fallacy	25.58%
Appeal to Emotion	11.37%
Faulty Generalization	10.18%
Fallacy of Credibility	9.90%
Ad Hominem	7.84%
Fallacy of Relevance	7.80%
Deductive Fallacy	6.50%
False Causality	5.11%
Fallacy of Extension	4.91%
Ad Populum	4.55%
False Dilemma	3.80%
Equivocation	1.94%
Circular Claim	0.51%

Table 4: Logical fallacy types and their frequencies in the LOGICCLIMATE dataset.

sentiment classification task), but more about the "form" or "structure" of the argument.

To advance the ability of models to detect fallacious logical structures, we draw inspirations from the history of logic (Russell, 2013). If we look into the time when Aristotle made his attempt to formulate a systematic study of logical, one of the most notable advancements is to move from contents to symbols, based on which Aristotle developed a system of rules (Gabbay and Woods, 2004). For example, he uses α, β, γ to distill arguments such as "Socrates is a man. All men are mortal. Therefore Socrates is mortal." into forms such as "α is a β. All β are γ. Therefore, α is γ.", where the variables act as placeholders. After establishing a system of valid and invalid argument structures, philosophers can refute a fallacious argument by comparing it to a list of fallacious logical forms (Aristotle, 2006).

Based on such inspirations, we propose a structure-aware classifier as a baseline model for our logical fallacy detection task. We first introduce a commonly used classification framework using pretrained models on natural language inference (NLI) in Section 3.1, and then we will propose our structure distillation process in Section 3.2.

3.1 Backbone: NLI-Based Classification with Pretrained Models

Motivated by the success of adapting NLI for classification tasks with unseen labels (Yin et al., 2019), we choose pretrained language models on NLI as the backbone of our logical fallacy classifier.

Specifically, a standard NLI-based pretrained language model for classification takes the sentence to classify as the *premise*. Then the model composes a *hypothesis* using the template of "This example is `[label name]`." The classifier checks whether the premise can entail the hypothesis. This NLI framework makes it easy for pretrained language models to adapt to unseen class labels such as our logical fallacy types.

3.2 Distilling Structure from Content

To build a model that encourages more attention to the *structure* of the text, we modify the premise and the hypothesis provided to the backbone NLI model (as shown in Figure 2): called the *structure-aware premise* and *structure-aware hypothesis*.

Structure-Aware Premise. Inspired by the process how ancient Greek philosophers refute an argument they have heard, we design an argument structure distiller by masking out content words in the premise (i.e., input text) and outputting a logical form with placeholders. In the example in Figure 2, "Jack is *a* good athlete. Jack comes from Canada. Therefore, *all* Canadians are good athletes.", we want the model to pay more attention to the structure as opposed to contents such as "good athletes." Thus, we build a distilled argument with placeholders "[MSK1] is a [MSK2]. [MSK1] comes from [MSK3]. Therefore, all [MSK3] are [MSK2]."

As shown in Figure 2, to distill the premise into the logical form, we identify all text spans that are paraphrases of each other and replace them with the same mask. Specifically, we first conduct coreference resolution using the `CoreNLP` package (Manning et al., 2014). Then, to identify word spans that are paraphrases of each other, we consider only non-stop words, lemmatize them via the `Stanza` package (Qi et al., 2020), and represent each word by its contextualized embedding generated by Sentence-BERT (Reimers and Gurevych, 2019), and calculate pair-wise cosine similarity. When the cosine similarity is larger than a threshold (by a grid search on the dev set), we identify the two words as similar. For illustration, we create a link between similar word pairs in Figure 2. When there are contiguous sequences of words that are linked to each other (e.g., "good athlete" and "good athletes"), we merge them and end up with two multi-word spans that are similar to each other. For each group i of similar text spans, we replace them with a mask token `[MSK`$_i$`]`.

Structure-Aware Hypothesis. NLI-based classification models (Yin et al., 2019) typically compose the hypothesis as a template sentence "This is an example of `[label name]`." However, in order to help our model perform a structure-aware matching of the logical fallacy instance, we also augment the hypothesis with the logical form for the logical fallacy type. For example, the logical form for *faulty generalization* in the example in Figure 2 is changed to: "[MSK1] has attribute [MSK2]. [MSK1] is a subset of [MSK3]. Therefore, all [MSK3] has attribute [MSK2]."

To look up the logical form of each fallacy, we refer to websites that introduce the logical fallacies, extract the expressions such as "Circular reasoning is often of the form: 'A is true because B is true; B is true because A is true."', and compile the logical forms using our masking format. We provide the list of logical forms in Appendix A.3.

4 Experiments

4.1 Experimental Setup

Evaluation Metrics. Since the nature of the logical fallacy detection task is a multi-label classification with class imbalance, we use *micro* F_1 as the main evaluation metric. Additionally, we also report precision, recall and accuracy.

Baselines. We test the performance of 12 existing large language models, including five zero-shot models and seven finetuned models. For zero-shot models, we use the zero-shot classifier by `transformers` Python package (Wolf et al., 2020) implemented using RoBERTa large (Liu et al., 2019) and BART large (Lewis et al., 2020) finetuned on the multi-genre natural language inference (MNLI) task (Williams et al., 2018). We also include the task-aware representation of sentences (TARS) (Halder et al., 2020a) provided by FLAIR (Akbik et al., 2019). Moreover, we also try directly using GPT-2 (Radford et al., 2019) and GPT-3 (Brown et al., 2020). For GPT-3, we designed a prompt for the auto-completion function to predict the label of text, and for GPT-2, we calculate the perplexity of every possible label with the text and choose the label with the lowest perplexity. See Appendix B.1 for more implementation details.

For finetuned baselines, we finetune seven commonly used pretrained language models on the LOGIC dataset, including ALBERT (Lan et al., 2020), BERT (Devlin et al., 2019), BigBird (Zaheer et al., 2020), DeBERTa (He et al., 2021), DistilBERT (Sanh et al., 2019), Electra (Clark et al., 2020), MobileBERT (Sun et al., 2020), and RoBERTa (Liu et al., 2019). See Appendix B.2 for implementation details.

Implementation Details. We describe the implementation details of our structure-aware classifier in Appendix B.3.

	F_1	P	R	Acc
Random	12.02	7.24	35.00	0.00
***Zero-shot classifiers directly tested on* LOGIC**				
TARS	8.62	3.86	6.67	2.33
BART-MNLI	11.05	6.63	33.67	0.00
GPT3	12.20	12.00	12.00	12.00
RoBERTa-MNLI	12.22	7.51	36.00	0.33
GPT2	13.67	13.67	13.67	13.67
***Finetuned and tested on* LOGIC**				
ALBERT	12.50	6.67	100.00	0.00
BigBird	15.02	8.61	90.00	0.33
DistilBERT	26.96	22.06	74.00	4.67
MobileBERT	35.68	29.05	71.00	7.33
BERT	45.80	40.73	73.67	18.00
DeBERTa	50.29	45.79	73.00	24.67
Electra	53.31	51.59	72.33	35.66
Electra-*StructAware*	**58.77**	55.25	63.67	47.67
Ablation study on the proposed model				
Raw Prem. + Str. Hypo.	56.72	54.87	76.67	37.67
Str. Prem. + Raw Hypo.	44.56	39.74	71.00	18.33

Table 5: Model performance on LOGIC by the ascending order of the main metric, micro F_1 (F_1). In addition, we also report the precision (P), recall (R), and accuracy (Acc). In the ablation study, we report the performance of two settings: a raw premise (i.e., keeping the original text input) with a structure-aware hypothesis (Raw Prem. + Str. Hypo.), and a structure-aware premise with a raw hypothesis (Str. Prem. + Raw Hypo.).

	F_1	P	R	Freq.
Faulty Generalization	60.24	47.62	81.97	18.01
Ad Hominem	78.65	72.92	85.37	12.33
Ad Populum	79.45	67.44	96.67	9.47
False Causality	58.82	62.50	55.56	8.82
Circular Claim	46.43	35.14	68.42	6.98
Appeal to Emotion	50.00	48.00	52.17	6.82
Fallacy of Relevance	39.22	37.04	41.67	6.61
Deductive Fallacy	25.81	16.67	57.14	6.21
Intentional Fallacy	26.23	17.39	53.33	5.84
Fallacy of Extension	49.18	37.50	71.43	5.76
False Dilemma	55.00	39.29	91.67	5.76
Fallacy of Credibility	58.82	58.82	58.82	5.39
Equivocation	33.33	100.00	20.00	2.00
Overall	58.77	55.25	63.67	100

Table 6: Class-specific performance achieved by Electra-*StructAware*. For each class, we report the F_1 score, precision (P), recall (R), and the frequency (Freq.) of the class in the LOGIC dataset. Note that the Freq. column is copied from Table 1.

4.2 Main Results

We test how well existing language models can address the task of logical fallacy classification, and check whether our proposed model can lead to performance improvement.

Zero-Shot Classifiers. In Table 5, we first look into some commonly used off-the-shelf zero-shot classification models. Surprisingly, most zero-shot classifiers are not much better than randomly choosing a label (i.e., the "Random" baseline in Table 5). The RoBERTa-MNLI classifier and GPT2, which achieve 12.22% and 13.67% F_1 scores, respectively are only marginally better than random guessing.

Finetuned Models. We further look into the effectiveness of finetuned large language models. The model performance is shown in ascending order in Table 5. According to our main metric, F_1, the best language model is Electra, which achieves 53.31% F_1 scores, followed by DeBERTa which achieves 50.29%.

Then, we adopt Electra as the backbone model to test our proposed structure-aware classifier (denoted as Electra-*StructAware*). Our model outperforms Electra by 5.46%, which is a fairly large margin. This implies the importance of encouraging the model to shift its attention to the logical form. Our model also achieves the highest exact match result, 47.67%, which is 12.01% better than the best performance among all language models finetuned in the standard way.

Ablation Study. Through the ablation study in Table 5, we can see that raw premise (i.e., keeping the original text input) with structure-aware hypothesis yields 56.72%, which can be attributed to the fact that the logical form provides richer information than just the label name. On the contrary, the structure-aware premise with the raw hypothesis of just the label name leads to a much worse result, perhaps because the model cannot easily figure out the correspondence between the masked text input and the label name. The ablation study also demonstrates that the best performance of our model comes from the matching between the logical form and the masked text input.

4.3 Class-Specific Performance

In addition to the overall performance of our proposed Electra-*StructAware* model, we further analyze its class-specific performance in Table 6.

Many of the logical fallacy classes can reach F_1 scores close to the overall F_1 of 58.77%. However, there are some logical fallacy types with relatively higher or lower performance. As the prediction